

A complete Mittag–Leffler classification for $\mathcal{H}(K_p + K_q)$

Candidate full-solution packet for arXiv:1703.02792, Question 3.10(ii)

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Status: candidate full solution, likely valid, subject to human review. The operator question is classified exactly by the standard zero-free phase boundary for the two-parameter Mittag–Leffler function.

1 Source question and answer

For $a > 0$, Anand and Chavan define

$$K_a(z, w) = \sum_{k \geq 0} (k+1)^a (z\bar{w})^k, \quad z, w \in \mathbb{D}.$$

Question 3.10(ii) of *On sum of two subnormal kernels*, arXiv:1703.02792, asks for a necessary and sufficient condition for M_z on $\mathcal{H}(K_p + K_q)$ to be subnormal [1]. Their paper proves subnormality when $0 < p \leq q \leq p+1$, and proves for a gap of exactly two that $\mathcal{H}(K_p + K_{p+2})$ is subnormal exactly when $p \geq 1$.

Write

$$m = \min\{p, q\}, \quad M = \max\{p, q\}, \quad r = M - m.$$

Let $E_{\alpha, \beta}$ denote the two-parameter Mittag–Leffler function

$$E_{\alpha, \beta}(z) = \sum_{j=0}^{\infty} \frac{z^j}{\Gamma(\alpha j + \beta)}.$$

For $1 < r \leq 2$, let $\phi(r)$ be the zero-free phase boundary on the negative real axis: it is the least β such that

$$E_{r, \beta'}(-x) \geq 0 \quad \text{for every } x \geq 0 \text{ and every } \beta' \geq \beta. \quad (1)$$

This is the standard α – β boundary tabulated by Hanneken, Achar, and Vaught [2]; a recent primary-source computation reproduces the boundary and its values [3]. In particular, $\phi(2) = 3$.

Theorem 1 (Complete classification). *For $p, q > 0$, the multiplication operator M_z on $\mathcal{H}(K_p + K_q)$ is subnormal precisely in the following cases:*

1. $r = 0$;
2. $0 < r \leq 1$;
3. $1 < r \leq 2$ and $M \geq \phi(r)$.

It is never subnormal when $r > 2$.

Thus the answer is elementary outside the transition strip $1 < r < 2$, and in that strip the exact threshold is the already established Mittag–Leffler zero-free curve. Some representative values are

r	1.05	1.20	1.50	1.80	1.95	2.00
$\phi(r)$	1.05924	1.26674	1.79365	2.46779	2.86070	3.00000
$\phi(r) - r$	0.00924	0.06674	0.29365	0.66779	0.91070	1.00000

The last row is the cutoff for the smaller exponent m . At $r = 2$, the condition $M \geq 3$ is exactly $m \geq 1$, recovering the source theorem.

2 The discrete-to-continuous rigidity lemma

The source reduces subnormality to the Hausdorff moment property of

$$a_k = \frac{1}{(k+1)^m + (k+1)^M}, \quad k \geq 0. \quad (1)$$

The central point is that, for this particular sequence, sampling only at the positive integers loses no information.

Lemma 1 (Rigidity of the Hausdorff interpolation). *Put*

$$f(s) = \frac{1}{s^m + s^M} = \frac{s^{-m}}{1 + s^r} \quad (s > 0),$$

using the principal powers. Then (1) is a Hausdorff moment sequence if and only if the inverse Laplace density

$$h_{m,M}(t) = t^{M-1} E_{r,M}(-t^r), \quad t > 0, \quad (2)$$

is nonnegative.

Proof. The standard transform identity gives

$$\mathcal{L}\{t^{M-1} E_{r,M}(-t^r)\}(s) = \frac{s^{r-M}}{1 + s^r} = \frac{s^{-m}}{1 + s^r} = f(s). \quad (3)$$

If $h_{m,M} \geq 0$, then

$$a_k = f(k+1) = \int_0^\infty e^{-(k+1)t} h_{m,M}(t) dt = \int_0^1 x^k h_{m,M}(-\log x) dx,$$

so (a_k) is a Hausdorff moment sequence.

Conversely, suppose $a_k = \int_{[0,1]} x^k d\mu(x)$ for a finite positive measure μ . On the right half-plane put

$$F(z) = \int_{[0,1]} x^z d\mu(x), \quad G(z) = f(z+1).$$

Both are bounded and analytic for $\Re z > 0$. For G , note that every principal-branch solution of $s^r = -1$ has modulus one and real part strictly less than one, whereas $s = z+1$ has real part greater than one. The function $F - G$ vanishes at every positive integer. A nonzero bounded analytic function on a half-plane has zeros satisfying the Blaschke condition

$$\sum_n \frac{\Re z_n}{1 + |z_n|^2} < \infty;$$

the positive integers violate it. Hence $F = G$ identically.

Push $\mu|_{[0,1]}$ forward under $x = e^{-t}$. The equality $F = G$, together with (3), says that its Laplace transform is also the transform of the finite signed measure

$$e^{-t}h_{m,M}(t) dt.$$

This signed measure is finite: the poles of f all have real part strictly less than one, and the branch-cut contribution has only algebraic growth. Uniqueness of Laplace transforms of finite measures therefore identifies the two measures. Since the first is positive, $h_{m,M} \geq 0$ almost everywhere, and continuity upgrades this to pointwise nonnegativity. $\square$

The bounded-analytic step is essential. Complete monotonicity of the continuous function f would certainly suffice, but without the rigidity lemma it would not automatically be necessary for an integer-sampled Hausdorff sequence.

3 Proof of the classification

Proof of Theorem 1. If $r = 0$, then

$$a_k = \frac{1}{2(k+1)^m} = \frac{1}{2\Gamma(m)} \int_0^1 x^k (-\log x)^{m-1} dx,$$

so M_z is subnormal.

Assume first that $0 < r \leq 1$. Since $M = m + r > r$, Schneider's sharp theorem says that $E_{r,M}(-x)$ is completely monotone, hence nonnegative, on $[0, \infty)$ [4, 5]. Lemma 1 gives subnormality for every $m > 0$. This recovers and slightly reorganizes the source's gap-at-most-one region.

Now let $1 < r \leq 2$. By Lemma 1, subnormality is equivalent to

$$E_{r,M}(-x) \geq 0 \quad (x \geq 0).$$

The positivity set is an upper interval in the second parameter. Indeed, for $\delta > 0$ the fractional-integration identity is

$$t^{M+\delta-1} E_{r,M+\delta}(-t^r) = \frac{1}{\Gamma(\delta)} \int_0^t (t-u)^{\delta-1} u^{M-1} E_{r,M}(-u^r) du.$$

Consequently, the exact threshold is the zero-free boundary $M = \phi(r)$ defined in (1). This proves the third case, including its endpoint. The identity $\phi(2) = 3$ gives precisely the source's result for $K_m + K_{m+2}$.

Finally suppose $r > 2$. The principal-branch denominator $1 + s^r$ has a dominant conjugate pair of simple poles

$$\zeta_{\pm} = e^{\pm i\pi/r}, \quad \Re \zeta_{\pm} = \cos(\pi/r) > 0.$$

Moving the inverse-Laplace contour left and collecting residues yields

$$h_{m,M}(t) = \frac{2}{r} e^{t \cos(\pi/r)} \cos\left(t \sin(\pi/r) + \frac{(1-M)\pi}{r}\right) + o\left(e^{t \cos(\pi/r)}\right). \quad (4)$$

All other poles have smaller real part, and the branch-cut term is subexponential. The leading cosine changes sign infinitely often. Thus $h_{m,M}$ is not nonnegative, and Lemma 1 rules out subnormality for every $m > 0$. $\square$

4 Checks, novelty, and limitations

The bundled script evaluates the defining Mittag–Leffler series at high precision. It confirms sign changes just below representative tabulated phase thresholds, positivity at sampled points just above them, the identity $h_{1,3}(t) = 1 - \cos t$ at $r = 2$, and the oscillatory obstruction for $r > 2$. These computations are sanity checks; the proof rests on the transform, Blaschke, phase-boundary, and residue arguments.

The initially plausible elementary cutoff $m \geq r - 1$ is not sharp. For example, at $r = 1.5$ the true cutoff is $m \geq \phi(1.5) - 1.5 \approx 0.29365$, whereas $r - 1 = 0.5$. The phase-boundary formulation is therefore mathematically substantive, not cosmetic.

On 11 August 2026, the run registry and cheap indexes were searched for arXiv:1703.02792, the exact kernel notation, and the moment sequence (1). Bounded primary-source searches for the title, exact question, and close combinations of “subnormal kernel”, “Hausdorff moment”, and “Mittag–Leffler” found the source paper and the independent special-function phase literature, but no later paper making this operator-theoretic connection or answering Question 3.10(ii). The novelty is the exact reduction and synthesis; the special-function phase curve itself is not claimed as new.

Human review should focus on (i) boundedness and zero-set uniqueness in Lemma 1; (ii) uniqueness of the signed inverse-Laplace measure after the unit shift; and (iii) the interpretation and endpoint of the published zero-free boundary ϕ . The result answers Question 3.10(ii) only; Questions 3.10(i) and 3.11 remain outside its scope.

A Source-page evidence

Figure 1 reproduces printed page 12 of the source PDF, including Question 3.10(ii).

References

- [1] A. Anand and S. Chavan, *On sum of two subnormal kernels*, arXiv:1703.02792v2 [math.FA], 2017.
- [2] J. W. Hanneken, B. N. Achar, and D. M. Vaught, *An alpha–beta phase diagram representation of the zeros and properties of the Mittag–Leffler function*, Advances in Mathematical Physics (2013), Article ID 421685.
- [3] A. H. Honain, K. M. Furati, I. O. Sarumi, and A. Q. M. Khaliq, *Rational approximations for oscillatory two-parameter Mittag–Leffler function*, arXiv:2312.07444, 2023; Fractal and Fractional 8 (2024), 319.
- [4] W. R. Schneider, *Completely monotone generalized Mittag–Leffler functions*, Expositiones Mathematicae 14 (1996), 3–16.
- [5] T. Simon, *Mittag–Leffler functions and complete monotonicity*, arXiv:1312.4513, 2013.

operator $M_{z,\lambda}$ on $H(K_\lambda)$ is a subnormal contraction if and only if $\lambda \geq d$. The same can also be verified by using Lemma 3.1 and part (i) of Theorem 2.11. Similarly, by using Lemma 3.1 and part (ii) of Theorem 2.11 one may conclude that the Cauchy dual tuple $M_{z,\lambda}^\sharp$ is a joint complete hyperexpansion if and only if $d \leq \lambda \leq d+1$. Thus, if we choose λ and λ' are such that $d \leq \lambda, \lambda' \leq d+1$. Then K_λ and $K_{\lambda'} \in \mathcal{K}_\sigma$. It now follows from Theorem 3.7 that the multiplication operator M_z on $\mathcal{H}(K_\lambda + K_{\lambda'})$ is subnormal. This is also included in the following example.

Example 3.8. Let $0 < d \leq \lambda' \leq \lambda \leq \lambda' + 1$. Note that the norm on $\mathcal{H}(K_\lambda + K_{\lambda'})$ admits the slice representation $[\sigma/2, H^2(\gamma)]$, where $\gamma_k^2 = \frac{2(d)_k}{(\lambda)_k + (\lambda')_k}$ for all $k \in \mathbb{Z}_+$. From the proof of Proposition 2.12 it is clear that $\{\gamma_k^2\}_{k \in \mathbb{Z}_+}$ is completely monotone. Hence, the multiplication operator M_z on $\mathcal{H}(K_\lambda + K_{\lambda'})$ is subnormal.

A d -tuple $S = (S_1, \dots, S_d)$ of commuting bounded linear operators $S_1, \dots, S_d$ in $\mathcal{B}(\mathcal{H})$ is a spherical isometry if $S_1^* S_1 + \dots + S_d^* S_d = I$. In other words, $Q_S(I) = I$. The most interesting example of a spherical isometry is the Szegő d -shift; that is, the d -tuple M_z of multiplication operators $M_{z_1}, \dots, M_{z_d}$ on the Hardy space $H^2(\partial\mathbb{B})$ of the unit ball.

Let ν be a Reinhardt measure. Consider the multiplication d -tuple M_z on a reproducing kernel Hilbert space $\mathcal{H}(K^\nu)$ determined by the reproducing kernel

$$(3.2) \quad K^\nu(z, w) = \sum_{\alpha \in \mathbb{Z}_+^d} \frac{z^\alpha \bar{w}^\alpha}{\|z^\alpha\|_{L^2(\partial\mathbb{B}, \nu)}^2} \quad (z, w \in \mathbb{B}).$$

Note that M_z is a spherical isometry. In this case, the norm on $\mathcal{H}(K^\nu)$ admits the slice representation $[\nu, H^2(\mathbb{D})]$, where $H^2(\mathbb{D})$ is the Hardy space of the unit disc.

Theorem 3.9. Let K^ν be the reproducing kernel given as in equation (3.2) and $\tilde{K}$ be any balanced kernel with the slice representation $[\nu, H^2(\tilde{\gamma})]$. Assume that the multiplication operator M_z on $\mathcal{H}(K^\nu + \tilde{K})$ is subnormal. Then the multiplication operator on $\mathcal{H}(\tilde{K})$ is subnormal.

Proof. Observe that the norm on $\mathcal{H}(K^\nu + \tilde{K})$ admits the slice representation $[\nu/2, H^2(\gamma)]$, where $\gamma_k^2 = 2(1 + 1/\tilde{\gamma}_k^2)^{-1}$ for all $k \in \mathbb{Z}_+$. Since M_z on $\mathcal{H}(K^\nu + \tilde{K})$ is subnormal, it follows from Lemma 3.1 that $\{\gamma_k^2\}_{k \in \mathbb{Z}_+}$ is a completely monotone sequence. Hence, $\{(1 + 1/\tilde{\gamma}_k^2)^{-1}\}_{k \in \mathbb{Z}_+}$ is a completely monotone sequence. If we replace a_k by $1/\tilde{\gamma}_k^2$ in the proof of the Proposition 2.9 we get that $\{\tilde{\gamma}_k^2\}_{k \in \mathbb{Z}_+}$ is completely monotone. Now, by applying Lemma 3.1 we conclude that the multiplication operator on $\mathcal{H}(\tilde{K})$ is subnormal. $\square$

We conclude the paper with the following questions:

Question 3.10. In view of Proposition 2.12 and Theorem 2.16, it is natural to ask that

- (i) what is the necessary and sufficient condition for the multiplication operator M_z on $\mathcal{H}(K_{\lambda,\mu} + K_{\lambda',\mu})$ to be subnormal?
- (ii) what is the necessary and sufficient condition for the multiplication operator M_z on $\mathcal{H}(K_p + K_q)$ to be subnormal?

Figure 1: Source paper, printed page 12: Question 3.10, including item (ii).